

Ishaan Fartyal

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Final-year BSc Mechanical Engineering student at TU/e with a focus on engineering analysis, data interpretation, and quantitative modelling for strategic business decisions. Experienced in using Python, financial and operational modelling, and cost analysis to turn data into insights for decision-making at high-tech companies.

EDUCATION

BSc Mechanical Engineering — Eindhoven University of Technology (TU/e) <i>Product Development & Entrepreneurship Elective Line (8 8 8)</i> Bachelor's Thesis: Determination of a Transition Plane in a Gas Atomization Simulation (Thermo-Fluids Engineering)	Expected Nov 2026 Eindhoven, Netherlands
IB Diploma (High School) — K. International School in Tokyo (KIST) 43/45 Physics HL 7 Mathematics AA HL 7 Economics SL 7	Jun 2021 Tokyo, Japan

EXPERIENCE

Cost Engineer — University Racing Eindhoven (URE) <i>Formula Student engineering team</i>	Sep 2025 – Aug 2026 Eindhoven, Netherlands
- Developed and maintained the car's detailed Bill of Materials (BOM) and contributed to the Costed-Carbonized BOM (CCBOM) for the engine and tractive system, covering component costs and lifecycle emissions across manufacturing and distribution. - Built an Excel and MATLAB cost model comparing prototype and mass-production scenarios for the inverter, evaluating in-house manufacturing and assembly versus outsourcing across production volumes to identify cost break-even points and sourcing implications. - Assessed cost and carbon implications of component and manufacturing choices, translating engineering decisions into financial and sustainability trade-offs. - Contributed the cost analysis that the team presented at Formula Student Czech Republic (11th of 52 in the Cost event) and Formula Student Germany.	
Mechanical Design Engineer — University Racing Eindhoven (URE) <i>Formula Student engineering team</i>	Sep 2024 – Aug 2025 Eindhoven, Netherlands
- Designed an in-wheel motor assembly for a race car using Siemens NX, translating design requirements, preferences, and constraints into a detailed CAD assembly. - Evaluated packaging, manufacturability and integration constraints with vehicle sub-teams, iterating the design around technical and production limitations.	

SELECTED WORK [↗](#)

Airbus A320 Ramp-Up Analysis ↗ <i>Independent Project</i>	Oct 2026
- Built a Python model of Airbus's A320 Family ramp-up to 75 aircraft a month, projecting yearly deliveries and cash flows to 2035 under limits on factory capacity, supplier availability and demand. - Valued the ramp-up at approximately €1.0bn NPV over ten years and found it breaks even only if suppliers support about 82% of the plan, with each point of availability worth €0.33bn. - Recommended timing investment to supplier readiness rather than a fixed schedule, as a two-year ramp-up delay turns NPV negative.	
ASML Technology Strategy ↗ <i>Independent Project</i>	Sep 2026
- Conducted an independent strategy analysis of ASML, combining industry structure, financial performance, technology trends, and supply-chain risk to assess the durability of its competitive advantage. - Evaluated strategic options around High-NA EUV commercialization, installed-base expansion, and selective vertical integration, developing a prioritized recommendation to sustain ASML's long-term market leadership.	
Finsight ↗ <i>Python-based Personal Finance Analytics App</i>	Sep 2026 – Present
Building a Python and Streamlit app that combines bank-statement CSVs from five bank formats into a dashboard of income, expenses, savings rate, recurring payments and monthly trends, with automated tests and a live online demo.	

ACTIVITIES

Japan U19 National Cricket Team — ICC Under-19 Cricket World Cup <i>International Athlete</i>	2020
Selected to represent Japan at the ICC Under-19 Cricket World Cup, competing in an international high-performance team environment.	
Student Athlete, Boxing/Kickboxing and Judo — E.S.K.B.V. Impact E.S.B.V. Samourais	2024 – 2026

SKILLS & ADDITIONAL INFORMATION

Technical Skills: Python (pandas, NumPy, Matplotlib, Plotly) Excel MATLAB Siemens NX Arduino	Languages: English & Hindi — Native Japanese — B2 Equivalent (JLPT N2) Dutch — Elementary (Actively learning)
Business Skills: Cost Modelling Financial Modelling (NPV) Scenario & Sensitivity Analysis Capacity & Make-or-Buy Analysis	Awards: Silver Award Scholarship (2021)
Work Authorization: Eligible for Dutch Orientation Year (Zoekjaar)	Certifications: Corporate Finance Foundations – CFI, Sep 2026
	Interests: Solo Travel Stargazing Philosophy (Stoicism)